Pre-Analysis Plan

#ManyDaughters crowdscience project

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1. Introduction

This pre-analysis plan outlines the analysis plan of the research team with the team ID which participates in the crowdscience project #ManyDaughters. The task is to test a specific set of hypotheses (outlined below) using the GSOEP.

2. Study Design

2.1. Hypotheses

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters)> (X for respondents without daughters).

H2. Having daughters leads to more left-leaning political preferences.
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters)> (Y for respondents without daughters).

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters)> (Z for respondents without daughters).

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).
That is, denoting X as a measure of support for gender equality, X for male respondents with daughters −X for male respondents without daughters >X for female respondents with daughters−X for female respondents without daughters.

We use for each hypothesis different subsamples of the GSOEP data sample because different variables of interest (gender attitudes, left-leaning political attitudes, and gender-equality behavior) in the way we operationalise them are only available for specific waves. For the hypotheses including a respective variable, we restrict the sample to those years where these variables of interest are included in the SOEP questionnaires.

We restrict all subsamples to the age range between 25 and 65. We exclude those who are likely still in education (<25) and those in retirement (>65).

Sample H1

- GSOEP 2012+2018 (years including variables for gender attitudes: pl/plh0298_v1, pl/plh0298_v2)
- all respondents in prime adulthood, age 25-65

Sample H2

- GSOEP 2005, 2009, 2014, 2019 (years including left vs right political attitudes: pl/plh0004)
- all respondents in prime adulthood, age 25-65

Sample H3

- GSOEP after reunification to include both West and East Germans: 1990–newest wave available (2022)
- Respondents in prime adulthood, age 25-65), who report to be in a cohabitating couple relationship

Sample H4

- GSOEP 2012+2018 (years including variables for gender attitudes pl/plh0298_v1, pl/plh0298_v2)
- all respondents in prime adulthood, age 25-65

2.2. Variables of Interest

IV for all hypotheses:

- dummy indicating whether a person in a specific year has a daughter
    - (1) identify when giving birth: biobirth$kidgeb01-kidgeb15
    - (2) identify sex of the kid: biobirth$kidsex01-kidsex15
    - dummy == 0: no children / only sons; dummy == 1 if at least one child is a daughter

Outcomes

- **DV H1:** gender egalitarian attitudes, measured as agreement to "Child Under 3 Yr. Suffers When Mother is at Work": pl/plh0302_v1, pl/plh0302_v2; 0 if slightly or completely agree; 1 if slightly or fully disagree
- **DV H2**: political orientation (lrscale): plh0004, continuous
- **DV H3**: housework done in hours per work day (harmonised), pl/pli0043, continuous
- **DV H4:** same DV as in H1

Controls

- sex of the respondent: pequiv/d11102ll (IV in H3)
- # of children in household: pequiv/d11107
- Employment status of respondent: pgen/emplst

2.3. Model specifications

- H1 pooled sample, LPM, dummy if multiple observations of the same individual, clustered SE on pid (personal identifier)
- H2 pooled sample, OLS, dummy if multiple observations of the same individual, clustered SE on pid
- H3 twoway FE including interaction sex_respondent * dummy_daughter, clustered SE on pid
    - assumption: on average, females do more household work than males as shown by past research (e.g., Nitsche & Grunow, 2016); we would therefore expect that the effect of transitioning into having a daughter on hours spent with household work is more positive for males than for females
    - we include as IV an interaction between sex_respondent * dummy_daughter
- H4 pooled sample, LPM, dummy if multiple observations of the same individual, clustered SE on pid
    - we include as IV an interaction between sex_respondent * dummy_daughter

Equations

- H1 (between estimates): y_genderattitudes_i,t = alpha_i,t + beta_daughter_dummy_i,t + vector of controls
- H2 (between estimates): y_lrscale_i,t = alpha_i,t + beta_daughter_dummy_i,t + vector of controls
- H3 (within estimates): y_hhwork_i,t = alpha_i,t + + beta1(daughter_dummy_i,t * beta_sex_i) + beta2(daughter_dummy_i,t * beta_sex_i) + vector of controls
- H4 (between estimates): y_genderattitudes_i,t = alpha_i,t + beta1(daughter_dummy_i,t * beta_sex_i) + beta2(daughter_dummy_i,t * beta_sex_i) + vector of controls
- All models include clustered SEs on individual level

Test-statistics

- each hypothesis is confirmed when statistically significant on $p<0.01$-level in the hypothesized direction
- substantially, we take the effect size into account

Reference

Nitsche, N. & Grunow, D. (2016): Housework over the course of relationships: Gender ideology, resources, and the division of housework from a growth curve perspective. *Advances in Life Course Research,* 29, 80-94. https://doi.org/10.1016/j.alcr.2016.02.001